Two axes of medial-axis reconstruction: compactness, and the Busemann–Hessian detection trichotomy*

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Abstract

For a closed set X in a complete Riemannian manifold the medial-axis *reconstruction* $\mathcal{R}_X = \bigcup_{a \in M_X} B(a, r(a))$ is the union of the open medial balls. Białożyt's Theorem 6 [1] characterises $\mathcal{R}_X$ for $X \subseteq \mathbb{R}^n$ as the convex hull, less X, together with the interiors of the *defective* supporting half-spaces — those whose contact set is non-convex. Across the companion notes this characterisation took two seemingly different faithful forms: in non-positive curvature an exact, *strict* description by horoballs with a Chebyshev/horoconvex defect [2, 3, 4], and in positive curvature its total collapse, $\mathcal{R}_X = M \setminus X$ [5]. We show these are governed by *two* orthogonal invariants, of which one — the sign of the Busemann Hessian — is the note's subject. The maximal free region carrying the reconstruction near a point is a sublevel set of a Busemann/distance function: a horoball below zero, a metric *ball* above. Two questions decide the theory, and they are independent.

- *Partial versus total* is fixed by **compactness**, not by curvature: reconstruction misses exactly the points whose ascending distance-flow escapes to an end with a single foot, so on a *compact* M the two-geodesic axis reconstructs everything, $\mathcal{R}^g_X = M \setminus X$ (Theorem 6.4, any curvature), while a non-compact one is in general partial. The two-*point* axis M_X of Białożyt is finer: even on a compact M it is governed by cut-shielding, total only under a cut-locus rigidity (the round sphere) and partial otherwise.
- *The detection flavour*, when partial, is fixed by the sign of $\operatorname{Hess} b$ — by curvature, through Hessian comparison — because it sets whether the maximal free *convex* region is unbounded (an *ideal* centre) or bounded (a *finite* one): $\sec \le 0$ ($\operatorname{Hess} b \succeq 0$, convex unbounded horoballs, ideal centre) makes detection a convexity test (Chebyshev/Motzkin/Carnot hull) with $\mathcal{R}_X \subsetneq M \setminus X$; $\sec \equiv 0$ ($\operatorname{Hess} b \equiv 0$, half-space, ideal centre at infinity) is the Motzkin knife-edge (Białożyt verbatim); $\sec > 0$ ($\operatorname{Hess} b \preceq 0$, no unbounded convex free region by Bonnet–Myers or a soul, finite centre) collapses detection to a contact count.

The two co-vary on the classical models ($\mathbb{H}^n$: non-compact, $\sec \le 0$; $\mathbb{S}^n$: compact, $\sec > 0$), which invites their fusion into a single "curvature trichotomy"; they *decouple* on an open

*Capstone of a series generalising Theorem 6 of [1]: [2] carried its sufficiency half to Hadamard manifolds, [3] proved the full characterisation in real hyperbolic space, [4] gave the faithful curvature-free characterisation for every Hadamard manifold and isolated the Gauss obstruction, and [5] crossed into positive curvature. This note unifies them: it identifies the invariant — the sign of the Busemann Hessian, i.e. the sign of the sectional curvature — that governs the *detection flavour*, separates it from the orthogonal *compactness* axis that governs partial-versus-total, and reads each companion result as one cell of the resulting 2×2. Drafted with the assistance of Claude (Anthropic); all arguments were re-derived and checked by hand, and the positive-curvature branch verified numerically (the convexity-radius threshold and Hessian sign-flip on $\mathbb{S}^2$; the cap's lone-antipode medial axis; the open-paraboloid partiality; and the prolate-ellipsoid shielding transition). Remaining errors are ours.

sec > 0 paraboloid (finite centres, yet *partial*) and on a compact hyperbolic manifold (sec < 0, yet *total*). The reconstructing ball above zero is moreover *not* convex — it wraps around past the convexity radius (radius up to the diameter), which is the mechanism of totality; the convexity-radius and soul bounds apply to the convex regions that host the *ideal* centre, hence to the detection flavour, not to the carrier. The honest structure is therefore a 2×2 — compactness $\times$ Hessian sign — with Białożyt's theorem the curvature knife-edge of the detection axis.

Contents

1 Introduction

One theorem, two faces

For a closed set X in a complete Riemannian manifold M write M_X for its *medial axis* — the points with two or more nearest points of X — and, for $a \in M_X$, $r(a) = \operatorname{dist}(a, X)$ for its reach. The *reconstruction* of X is the union of the open medial balls,

$$\mathcal{R}_X = \bigcup_{a \in M_X} B\big(a, r(a)\big) \;\subseteq\; M \setminus X.$$

Białożyt's Theorem 6 [1] characterises $\mathcal{R}_X$ for closed $X \subseteq \mathbb{R}^n$: with $\widehat{X} = \overline{\operatorname{conv}}\, X$,

$$\mathcal{R}_X = (\widehat{X} \setminus X) \;\cup\; \bigcup \big\{ \operatorname{int} H : \; H \text{ a supporting half-space, } X \cap \partial H \neq \operatorname{conv}(X \cap \partial H) \big\}. \quad (1)$$

The companion notes carried this statement across the curvature spectrum, and on its face the two ends look unrelated.

Below zero. In every Hadamard manifold (sec ≤ 0) the half-space becomes a horoball, conv becomes a horoconvex hull hconv built from the Busemann geometry of M, and (1) holds verbatim:

$$\mathcal{R}_X = (\widehat{X} \setminus X) \;\cup\; \bigcup \big\{ \operatorname{int} \mathcal{H} : \; X \cap \partial \mathcal{H} \neq \operatorname{hconv}(X \cap \partial \mathcal{H}) \big\} \quad (2)$$

[4, Thm. 3.3–3.4], with hconv = conv on flat horospheres (so (2) contains (1) and the hyperbolic theorem [3]) and the Carnot hull on homogeneous ones. The reconstruction is a *strict* subset of $M \setminus X$, and the unreconstructed points are exactly those whose nearest-point ray escapes to infinity carrying a single foot.

Above zero. On the round sphere ($\sec \equiv +1$) the same engine gives the opposite conclusion: for every closed $X \subseteq \mathbb{S}^n$ with $|X| \geq 2$,

$$\mathcal{R}_X = \mathbb{S}^n \setminus X \tag{3}$$

[5, Thm. 4.1] — reconstruction is *total*, the certificate (2) strictly undercounts, and the convexity defect that organises (1)–(2) has no analogue: a convex spherical cap reconstructs its entire complement [5, Prop. 5.1].

The invariant that unifies them

The purpose of this note is to show that (2) and (3) are governed by two *independent* invariants, and to separate them — because reading them as one curvature trichotomy, as a first pass invites, conflates two genuinely different mechanisms. Both proofs run the same machine. Outside the hull, the reconstruction of a point p is carried by a *maximal free ball* $B(a, r(a)) \ni p$ — the sublevel set $\{b_a \leq 0\}$ of the distance-Busemann $b_a = d(\cdot, a) - r(a)$ — whose limiting form is a horoball in $\mathbb{H}^n$ and an ordinary metric ball on $\mathbb{S}^n$. Two questions about it decide the theory, and they pull apart.

(A) Partial versus total: compactness. The ascending generalised-gradient flow of d_X from p either reaches a medial centre or escapes to an end of M with a single foot; the latter is possible iff M is non-compact. Hence **a compact M is totally reconstructed** (Theorem 6.4, *any* curvature), and a non-compact one is in general partial — the "escape to infinity" named above. This is a statement about ends, not curvature.

(B) Detection flavour: the sign of $\operatorname{Hess} b$. When reconstruction *is* partial, what must a contact set do to admit a medial centre? This is fixed by whether the maximal free *convex* region is unbounded or bounded — equivalently, by the sign of $\operatorname{Hess} b$, which equals the sign of sec (Hessian comparison):

	$\operatorname{Hess} b$	convex free region	centre / detection
$\sec \leq 0$	$\succeq 0$ (convex b)	convex, unbounded	ideal $\xi \in \partial_\infty M$ / Chebyshev hull
$\sec \equiv 0$	$\equiv 0$ (affine b)	half-space	∞ / Motzkin convexity
$\sec > 0$	$\preceq 0$ (concave b)	bounded	finite $a \in M$ / ≥ 2 contacts

In $\sec \leq 0$, convexity of b lets horoballs run to infinity, so their centre is an ideal point ξ; a finite reconstructing centre must *drift* toward ξ and is medial only when its foot is non-Chebyshev — a genuine convexity test (Motzkin on flat horospheres, Carnot on curved ones). In $\sec > 0$, concavity of b past the convexity radius forbids *unbounded* convex free regions (Bonnet–Myers, or a point soul), so there is no ideal centre to drift to; a finite equidistant centre is medial as soon as its boundary carries two contacts — no convexity test (Proposition 3.4). The flat case is the knife-edge: b affine, the convex region a half-space, the would-be finite circumcentre receded to infinity and the convexity radius diverged — the unique curvature where Białożyt's conv-certificate is exact.

These two axes are *orthogonal*. They co-vary on the classical models ($\mathbb{H}^n$ pairs non-compact with $\sec \leq 0$; $\mathbb{S}^n$ pairs compact with $\sec > 0$), which is why they look like one curvature trichotomy; but an open $\sec > 0$ paraboloid has finite centres yet is *partial* (§6), and a compact hyperbolic manifold has $\sec < 0$ yet is *total*. Two consequences sharpen the positive branch and correct the naive reading. First, above zero the reconstructing *ball* is in general *not* convex: it wraps around past the convexity radius (radius up to the diameter π), and that wrap-around — not a bounded convex region — is what makes reconstruction total (the convex free regions carry only the focal-pole part of §6). Second, the convexity-radius/Bonnet–Myers/soul apparatus bounds the convex regions that would host an *ideal* centre, so it controls the detection *flavour* (B), *not* partial-versus-total (A). The honest statement is a 2×2 (Table in §7), with the "trichotomy" naming its detection-flavour axis.

Organisation

Section 2 fixes the common objects — medial axis, Busemann functions, the free ball carrying reconstruction, the defect template. Section 3 is the conceptual core: two short theorems, one per axis — compactness governs partial-versus-total (Theorem 3.1), the Busemann Hessian governs the detection-flavour trichotomy (Theorem 3.2) — together with the finite-equidistant-centre principle that trivialises detection above zero. Sections 4–6 then read each branch as an instance, quoting the companion notes for the detailed reconstruction theorems and giving the mechanism in trichotomy language: the Hadamard branch (§4, ideal centres, the faithful hconv characterisation and its Carnot/Alexandrov frontier), the flat knife-edge (§5, Białożyt as the degenerate limit of both branches), and the positive branch (§6, finite centres, total reconstruction, the contact-count collapse, and what survives off constant curvature). Section 7 collects the unified picture, the phase transition at $\sec = 0$, and the open problems each branch bequeaths.

2 The common objects

Throughout M is a complete Riemannian manifold, $X \subseteq M$ is closed and nonempty, d is the geodesic distance, and $B(a, \rho) = \{x : d(x, a) < \rho\}$. We recall M_X, $r(a) = d(a, X)$ and $\mathcal{R}_X = \bigcup_{a \in M_X} B(a, r(a))$ from §1; the medial balls are free ($B(a, r(a)) \cap X = \varnothing$), so always $\mathcal{R}_X \subseteq M \setminus X$. Two structural objects underlie all three branches.

The carrier: a maximal free ball, and its Busemann function. The object that carries reconstruction is a *maximal free ball* $\mathcal{H} = B(a, r(a))$, the sublevel set $\{b_a \leq 0\}$ of the distance-Busemann $b_a = d(\cdot, a) - r(a)$; its boundary $\partial\mathcal{H} = S(a, r(a))$ is the *horosphere*, its *contact set* is $K = X \cap \partial\mathcal{H}$, and its *centre* is a. Two limiting regimes specialise it. In $\sec \leq 0$ one drives the centre to an ideal point $\xi \in \partial_\infty M$ along a ray γ, and $b_a \to b_\xi(x) = \lim_{t\to\infty}(d(x, \gamma(t)) - t)$ (a genuine *Busemann function*, 1-Lipschitz), whose sublevel sets are the *horoballs* at ξ; the centre is then ideal. In $\sec > 0$ the centre stays finite: on $\mathbb{S}^n$, $b_v = d(\cdot, v) - \frac{\pi}{2}$ is the signed distance to the equator $S(v, \frac{\pi}{2})$ of a pole v, with $\{b_v < 0\} = B(v, \frac{\pi}{2})$ the free hemisphere and v the finite centre [5, §2,§9]. A *ball* $B(a, r(a))$ carries reconstruction at *any* radius and is convex only for $r(a)$ below the convexity radius; separately we call a free $\{b \leq c\}$ a *maximal free convex region* when it is in addition geodesically convex and inclusion-maximal among such. The distinction is the crux of the positive branch (§3): the convex regions control *detection flavour*, but the carriers are the balls, which above zero wrap around past the convexity radius and need not be convex. (In $\sec \leq 0$ the two coincide — horoballs are convex — so the classical theory never had to separate them.)

The defect template. Every branch instantiates the same shape of theorem,

$$\mathcal{R}_X = (\widehat{X} \setminus X) \;\cup\; \bigcup\{\operatorname{int}\mathcal{H} :\; \mathcal{H} \text{ maximal free, } \textit{defective}\}, \tag{4}$$

where $\widehat{X}$ is the hull (the complement of the union of all open free convex regions) and "defective" is a condition on the contact set K alone, decidable from X and the geometry of M with M_X absent. The content of the trichotomy is what "defective" *is*, and whether (4) is a strict description or collapses to $M \setminus X$. We will see that this is decided entirely by the centre type of $\mathcal{H}$.

The reconstruction engine, curvature-free. Two inputs are common to all branches and use no curvature sign beyond what is named.

Theorem 2.1 (Hull inclusion and the swallow; [2, Thm. 3.1], [4, Lem. 2.2],[5, Lem. 3.1])**.** *For every complete M and closed X:*

(1) (Hull.) $\widehat{X} \setminus X \subseteq \mathcal{R}_X$.

(2) (Swallow.) *If a horoball $\mathcal{H}$ admits medial centres $a_k \in M_X$ drifting to its ideal centre ξ, then $\operatorname{int}\mathcal{H} \subseteq \mathcal{R}_X$, with $d(p, a_k) - r(a_k) \to b_\xi(p) < 0$ for $p \in \operatorname{int}\mathcal{H}$ — only horofunction convergence,* no curvature bound.

The swallow is the *unbounded*-centre mechanism. When the centre is finite there is no limit to take: a single medial ball $B(a, r(a))$ with $p \in B(a, r(a))$ already gives $p \in \mathcal{R}_X$, and the work is only to make a medial — which is Proposition 3.4, not a swallow. What the engine leaves open is exactly the certificate: *which* $\mathcal{H}$ admit medial centres, read from X. That detection problem, and whether the flow can escape at all, are governed by the two axes of the next section.

3 The two axes

We isolate the single comparison-geometric fact behind the whole theory. Recall the Hessian comparison theorem (e.g. [7, Ch. 6]): for $r(\cdot) = d(\cdot, o)$ and the model space form of curvature κ,

$$\sec \le \kappa \implies \operatorname{Hess} r \succeq \operatorname{Hess} r_\kappa, \qquad \sec \ge \kappa \implies \operatorname{Hess} r \preceq \operatorname{Hess} r_\kappa,$$

on the tangential subspace $(\nabla r)^\perp$, with the model $\operatorname{Hess} r_\kappa = \mathfrak{c}_\kappa(r)\,(g - dr \otimes dr)$ and $\mathfrak{c}_\kappa = \sqrt{\kappa}\cot(\sqrt{\kappa}\, r)$ for $\kappa > 0$, $= 1/r$ for $\kappa = 0$, $= \sqrt{-\kappa}\coth(\sqrt{-\kappa}\, r)$ for $\kappa < 0$. Passing to the Busemann limit $r \to \infty$ at $\kappa = 0$ sends the model term $1/r \to 0$, leaving the sign of $\operatorname{Hess} b$ equal to the sign forced on $\operatorname{Hess} r$ by the curvature sign.

The reconstruction of a point $p \in M \setminus \widehat{X}$ is carried by a maximal free ball $B(a, r(a)) \ni p$. Two independent features of M govern it, and we take them in turn.

Theorem 3.1 (Compactness controls partial versus total)**.** *Let M be complete and $X \subseteq M$ closed, $|X| \ge 2$. If M is* compact*, the two-geodesic axis reconstructs everything,*

$$\mathcal{R}^g_X = M \setminus X,$$

whatever the curvature (Theorem 6.4). If M is non-compact*, reconstruction is in general strict: the ascending flow of d_X from p can run to an end of M with a single foot.*

Proof. On compact M the semiconcave d_X has a finite-length ascending flow resting at a critical point with 0 in the convex hull of its directions to X, hence ≥ 2 minimising geodesics; that the flow retains p in its moving ball — the Lieutier swallow property, unconditional below the cut distance and otherwise as in Remark 6.5 — is assembled into $\mathcal{R}^g_X = M \setminus X$ by Theorem 6.4. On non-compact M the flow can ascend $d_X \to \infty$ along a ray with a single foot: the $\mathbb{R}^n$ rays past a segment, or the paraboloid of Remark 3.3. □

This axis is about *ends*, not curvature — it separates neither $\mathbb{H}^n$ from a paraboloid nor a compact hyperbolic manifold from $\mathbb{S}^n$. (Białożyt's two-*point* axis M_X is finer: cut-shielding keeps it partial even on some compact M, Remark 6.6.)

Theorem 3.2 (The Busemann Hessian controls the detection flavour)**.** *Where reconstruction is partial, the sign of* Hess b *— equal to the sign of* sec *by Hessian comparison — decides whether the maximal free* convex *region is unbounded, hence what its contact set must do to admit a medial centre:*

(i) sec ≤ 0 (Hess $b \succeq 0$)*: convex and* unbounded*, an* ideal *centre* $\xi \in \partial_\infty M$*; medial iff the contact set is non-Chebyshev* (*Proposition 3.5*).

(ii) sec $\equiv 0$ (Hess $b \equiv 0$)*: the affine knife-edge — a half-space, ideal centre at infinity, detection Motzkin's.*

(iii) sec > 0 (Hess $b \preceq 0$)*:* bounded*, a* finite *centre; medial as soon as it carries* ≥ 2 *contacts, irrespective of their position* (*Proposition 3.4*).

Proof. The sign of Hess b is Hessian comparison in the Busemann limit ($r \to \infty$ kills the $\kappa = 0$ model term $1/r$). For (i), Busemann functions on a Hadamard manifold are convex [6, II.8.24], so $\{b \leq c\}$ is convex and, containing the asymptotic ray, unbounded; a finite point sees $\partial\mathcal{H}$ at non-constant distance, so it is no circumcentre — whence the Chebyshev condition of §4. (ii) is the affine specialisation. For (iii), comparison with $\kappa = \sup \sec > 0$ makes $\mathfrak{c}_\kappa(r) = \sqrt{\kappa}\cot(\sqrt{\kappa}r)$ change sign at the convexity radius $\pi/(2\sqrt{\kappa})$, past which $\{b \leq c\}$ is non-convex; each of the finite convexity radius, Bonnet–Myers (under sec $\geq \kappa > 0$), and the point soul of an open sec > 0 manifold [8, 9] then bounds every convex free region, so no ideal centre exists. The carrier itself is a metric ball, to which Proposition 3.4 applies at any radius. □

The two are *independent*: curvature sign (Theorem 3.2) is silent on partial-versus-total, which is compactness (Theorem 3.1). They co-vary on $\{\mathbb{H}^n, \mathbb{S}^n\}$ but decouple on the paraboloid and the compact space forms.

Remark 3.3 (The carrier is a ball, not a convex region; and total is compactness)*.* Two cautions keep the positive branch honest; both were verified numerically.

The carrier need not be convex. On $\mathbb{S}^n$ the cap $X = \bar{B}(e, \beta)$ ($\beta < \frac{\pi}{2}$) is reconstructed entirely by the single ball $B(-e, \pi - \beta)$ of radius $\pi - \beta > \frac{\pi}{2}$ — *non-convex*, wrapping past the convexity radius. The maximal free *convex* regions here are hemispheres tangent to the cap, meeting X in one point, so their poles are not medial (Proposition 3.4) and they reconstruct nothing: the convex-region carrier is *empty*, and totality comes from the wrap-around ball alone (the cap's medial axis is the lone antipode $-e$). So Theorem 3.2(iii) bounds the convex regions — killing the ideal centre, fixing the *flavour* — but the convexity radius does *not* bound the carriers, which reach radius up to the diameter π; the wrap-around is the mechanism of totality. This is Proposition 6.2 (the template undercounts) seen in advance.

Total is compactness, not $\sec > 0$. A paraboloid of revolution has $\sec > 0$ everywhere but is non-compact, with an end on which $\sec \to 0$. For X a small circle around the vertex, the vertex is equidistant from X and medial (its feet the whole circle), while every *other* interior point has a unique same-meridian foot; so $M_X = \{\text{vertex}\}$ and $\mathcal{R}_X$ is the disk it bounds. Outside, the meridian through a point is a *ray* — the flattening end carries no conjugate point along it — so it minimises from its foot forever, $d_X \to \infty$ with that single foot, and the point never reaches a medial centre: *partial*, with finite centres throughout (no ideal boundary). Numerically, d_X is unbounded up the axis and no point above the inner disk is reconstructed, the vertex ball reaching only the disk. Dually, a compact hyperbolic manifold has $\sec < 0$ yet is *total* for $\mathcal{R}_X^g$ (Theorem 6.4). Curvature sign and compactness, fused on $\{\mathbb{H}^n, \mathbb{S}^n\}$, are independent in general.

Behind Theorem 3.2 are two principles — the contrast in *detection* that the centre type forces. We state them here; each is proved in its branch.

Proposition 3.4 (Finite equidistant centre trivialises detection — any manifold)**.** *A free geodesic ball $B(a, \rho)$ with $\rho = d(a, X)$ has* every *boundary point at distance ρ from a. Hence if the contact set $K = X \cap \partial B(a, \rho)$ has two or more points, then $a \in M_X$ and $B(a, \rho) \subseteq \mathcal{R}_X$ — with no condition on the position or geometry of the contacts. A finite equidistant centre never asks "which contact is nearest," the question whose answer is the Chebyshev structure, so that structure plays no role.*

Proof. For $x \in \partial B(a, \rho)$, $d(a, x) = \rho = d(a, X)$, so every contact is a foot; two contacts give two feet, $a \in M_X$ with $r(a) = \rho$, and every point of $B(a, \rho)$ lies in this medial ball [5, Lem. 3.1, Prop. 3.3]. □

Proposition 3.5 (Ideal centre demands a convexity test — $\sec \leq 0$)**.** *Let $\mathcal{H}$ be a maximal free horoball at ξ with contact set K on its horosphere. A finite centre high over a horosphere point σ is medial iff σ has two or more nearest contacts in the intrinsic geometry of $\partial\mathcal{H}$; and a drift to ξ reconstructing* $\operatorname{int}\mathcal{H}$ *exists iff $K \neq \operatorname{hconv}(K)$. Detection is the horoconvex-hull (Chebyshev/Motzkin) question.*

Proof. This is the deepest-shadow necessity and the swallow sufficiency of [4, Thm. 3.3]; see §4. The point is that, $\mathcal{H}$ being unbounded, the centre is *not* equidistant from $\partial\mathcal{H}$ (Theorem 3.2(i)), so being medial is a genuine nearest-among-many condition — non-Chebyshev — rather than the trivial count of Proposition 3.4. □

Remark 3.6 (The Hessian sign-flip, concretely and numerically). On $\mathbb{S}^n$ the contrast is visible in one formula. For $b_v(x) = d(x, v) - \frac{\pi}{2}$,

$$\operatorname{Hess} b_v = \cot\big(d(x, v)\big)\,\big(g - dr \otimes dr\big),$$

positive inside the v-hemisphere ($d < \frac{\pi}{2}$, where the would-be horoball is convex), vanishing on the equator ($d = \frac{\pi}{2}$), and *negative* outside ($d > \frac{\pi}{2}$): the signed-distance "Busemann" function is concave past the convexity radius, so the maximal free convex region is exactly the hemisphere $B(v, \frac{\pi}{2})$ — bounded, finite centre v. Both facts were checked numerically on $\mathbb{S}^2$: a cap of angular radius R is geodesically convex iff $R \leq \frac{\pi}{2}$ (convex at $R = 1.551$, leaking at $R = 1.591$), and the tangential eigenvalue of $\operatorname{Hess} d(\cdot, v)$ matches $\cot r$ to four places, flipping sign exactly at $r = \frac{\pi}{2}$. This is the $\kappa = +1$ instance of Theorem 3.2(iii).

Remark 3.7 (Why the flat case is a knife-edge). The two branches meet only at $\sec \equiv 0$, and degenerately. View $\mathbb{S}^n_R$ of radius R (curvature $R^{-2} > 0$): the convexity radius is $\frac{\pi R}{2} \to \infty$ and the focal circumcentre of a free hemisphere recedes to distance $\frac{\pi R}{2} \to \infty$ as $R \to \infty$ [5, Rem. 3.2], so the finite centre escapes to infinity and the bounded ball opens into a half-space — exactly the $\sec = 0$ picture of (ii). Symmetrically, sharpening $\sec \le -a^2 < 0$ contracts horoballs toward ξ; relaxing $a \to 0$ opens them to half-spaces. The flat case is the unique fixed point of both limits: $\operatorname{Hess} b \equiv 0$, neither convex nor concave, the ideal centre sitting at the finite/infinite boundary. It is no accident that Białożyt's theorem — the one case with a clean conv certificate that is simultaneously strict and Motzkin-detected — lives precisely there.

4 The branch $\sec \le 0$: ideal centres and the Chebyshev defect

In non-positive curvature Theorem 3.2(i) gives unbounded horoballs with ideal centres, and Proposition 3.5 makes detection a convexity question. The faithful characterisation is the template (2), proved in [4] and quoted here in trichotomy form.

Definition 4.1 (Horospherical convex hull; [4, Def. 3.1])**.** *For a maximal free horoball $\mathcal{H}$ at ξ with horosphere $S = \partial\mathcal{H}$ and an ideal direction u, let $L_u(y) = \frac{d}{d\varepsilon}\big|_{0^+} b_{\xi_\varepsilon}(y)$ be the one-sided derivative of the Busemann function along a tilt $\xi_\varepsilon \to \xi$. The* horospherical convex hull *of $K \subseteq S$ is*

$$\operatorname{hconv}(K) = \big\{\sigma \in S :\; L_u(\sigma) \ge \inf\nolimits_K L_u \textit{ for every } u\big\},$$

the trace on S of the intersection of the tilted horoballs containing K. It is a genuine hull, built from the Busemann geometry of M alone, equal to conv *on a flat horosphere and to the Carnot hull on a homogeneous one.*

Theorem 4.2 (Faithful characterisation in $\sec \le 0$; [4, Thm. 3.3–3.4])**.** *For every Hadamard manifold M and every closed $X \subseteq M$,*

$$\mathcal{R}_X = (\widehat{X} \setminus X) \;\cup\; \bigcup\big\{\operatorname{int}\mathcal{H} :\; X \cap \partial\mathcal{H} \ne \operatorname{hconv}(X \cap \partial\mathcal{H})\big\},$$

the defect certificate — the contact set is not horospherically convex — being decidable from X and the geometry of M, with M_X absent. It reduces to $X \cap \partial\mathcal{H} \ne \operatorname{conv}(\cdot)$ on flat horospheres ($\mathbb{R}^n$, $\mathbb{H}^n$) and to its Carnot form on homogeneous ones. The reconstruction is strict: $\mathcal{R}_X \subsetneq M \setminus X$ whenever X has a point whose nearest-point ray escapes to infinity with a single foot.

The two analytic inputs are both curvature-free and both visible in the trichotomy. *Necessity* is Danskin's first-order condition for the depth $D(\xi) = \inf_x b_\xi(x) - b_\xi(p)$: the deepest free horoball over p has its foot $\sigma(p)$ in $\operatorname{hconv}(K) \setminus K$, so $K \ne \operatorname{hconv}(K)$ [4, Thm. 3.3]. *Sufficiency* is the swallow (Theorem 2.1(2)) along the medial centres drifting to ξ — which exist precisely because, the centre being ideal (Theorem 3.2(i)), a non-horoconvex contact set surrounds the foot on every tilt and blocks ray-escape, so the trichotomy of point types [4, Thm. 2.1(4)] forces drift.

The Gauss obstruction and the detection frontier. Reading the ambient certificate $K \ne \operatorname{hconv}(K)$ as an *intrinsic* convexity of K in $\partial\mathcal{H}$ — the form Theorem 6 takes — requires the horosphere geometry, governed by the Gauss equation $K_S = K_M + \det \mathrm{II}$ [4, Prop. 4.1]. This is where the branch develops its own hierarchy, all of it on the *ideal-centre* side of the trichotomy:

- *Flat horospheres* ($K_S \equiv 0$: $\mathbb{H}^n$, and the warped products M_f of variable pinched curvature). Motzkin's theorem applies on the flat horosphere and detection is exactly non-convexity of K — the first reconstruction theorems beyond constant curvature [4, Thm. 7.2].

- CAT(0) *horospheres* (warped over a Hadamard fibre). The sharpness half (convex $\Rightarrow$ Chebyshev) is unconditional; detection reduces to the Motzkin property of the fibre [4, Thm. 8.3].

- *Positively curved horospheres* (the Heisenberg horosphere of $\mathbb{CH}^2$). Here K_S takes both signs; localisation is carried out exactly in the Carnot metric and detection is *graded non-Chebyshev* — the first reconstruction on a non-CAT(0) horosphere [4, Thm. 9.2].

- *The frontier* ($\mathbb{CH}^n$, $n \geq 3$; pinched sec $\leq -a^2$). Detection is a Chebyshev-set problem in a Carnot group, or Motzkin's theorem in an Alexandrov space of indefinite curvature — open [4, §10].

The entire frontier is a sec ≤ 0 phenomenon: it lives on the ideal-centre branch, where "defective" is a convexity and the open question is *which* convexity. As §6 will show, it has no counterpart above zero.

5 The knife-edge sec $\equiv 0$: Białożyt's theorem

The flat case is the degenerate boundary of Theorem 3.2(ii), where both descriptions of the centre coincide at infinity (Remark 3.7). It is worth recording exactly what each branch's mechanism becomes there, because it shows Białożyt's theorem (1) as the simultaneous limit of both.

In $\mathbb{R}^n$ the Busemann function $b_u(x) = -\langle x, u\rangle$ is affine (Theorem 3.2(ii)), the maximal free convex region is a half-space H, and:

- *From the ideal-centre side:* hconv = conv (the tilts $L_u(y) = \langle y, u^\perp\rangle$ are linear, Definition 4.1), so the defect $K \neq \mathrm{hconv}(K)$ is $K \neq \mathrm{conv}(K)$ — Motzkin's non-convexity [12]. The reconstruction is strict: a point on the line through two isolated contacts, beyond their segment, sends its nearest-point ray to infinity with a single foot and is never reconstructed.

- *From the finite-centre side:* the circumcentre of a free ball through two contacts sits at the focal point, which in $\mathbb{R}^n$ has receded to infinity (Remark 3.7); a centre at finite height over a *convex* contact set has a unique foot (convex $\Rightarrow$ Chebyshev), so it is not medial — which is the same Motzkin condition, seen as the failure of Proposition 3.4 when the equidistant centre is ideal.

Theorem 5.1 (Białożyt; [1, Thm. 6]). *For closed $X \subseteq \mathbb{R}^n$, (1) holds: $\mathcal{R}_X$ is the convex hull less X, together with the interiors of the supporting half-spaces whose contact set is non-convex.*

This is the unique curvature on which the certificate is at once *exact*, *strict*, and phrased by ordinary conv. Move down to sec < 0 and conv must become hconv to stay faithful (§4); move up to sec > 0 and the certificate, still formally available, stops being exact and is overwhelmed by total reconstruction (§6). The knife-edge is the only place the nineteenth-century statement is the whole truth.

6 The branch $\sec > 0$: finite centres and total reconstruction

Above zero, Theorem 3.2(iii) leaves no ideal centre: every centre is finite, and Proposition 3.4 collapses detection to a contact count. The *carriers*, though, are balls of any radius up to the diameter — convex only up to the convexity radius, then wrapping around (Remark 3.3) — and it is the wrap-around, together with compactness (Theorem 3.1), that on the round sphere makes the collapse *total*.

Theorem 6.1 (Total reconstruction on $\mathbb{S}^n$; [5, Thm. 4.1]). *For every closed $X \subseteq \mathbb{S}^n$ with $|X| \geq 2$,*

$$\mathcal{R}_X = \mathbb{S}^n \setminus X.$$

The mechanism is the finite centre throughout. The sharpest case is the *focal pole*: a free hemisphere H_v° whose equator meets X in ≥ 2 points has, at its pole v, a finite centre equidistant at $\frac{\pi}{2}$ from the entire equator, so $v \in M_X$ and $H_v^\circ \subseteq \mathcal{R}_X$ *regardless of the contact set's shape* (Proposition 3.4 with $\rho = \frac{\pi}{2}$) — there is no Motzkin step, no Chebyshev problem, the equatorial geometry idle. Globally, the proof follows the nearest-point ray of any $p \notin M_X$ from its foot x_0: $\mathbb{S}^n$ being compact (Theorem 3.1), the ray cannot escape, and the antipode being the farthest point at distance π, x_0 is overtaken by a rival contact before it — producing a finite medial centre that still contains p. The generalised-gradient flow of d_X [10, 11] makes this unconditional: on the compact $\mathbb{S}^n$ it has finite length, rests only at critical points, and a critical point of reach $< \pi$ carries two distinct nearest points [5, Thm. 4.1, Step 3].

Proposition 6.2 (The certificate collapses; [5, Prop. 5.1–5.2]). *The naive template* (4) *with "defective = non-convex contact set" strictly undercounts on $\mathbb{S}^n$. A closed spherical cap $X = \bar{B}(e, \beta)$, $\beta < \frac{\pi}{2}$, is convex with smooth strictly convex boundary, so no supporting hemisphere is defective and the template predicts $\varnothing$ — yet $\mathcal{R}_X = \mathbb{S}^n \setminus X$, the whole complement reconstructed from the single focal centre $-e$. A latitude circle likewise has every supporting hemisphere touching once, no defective facet, yet its far hemisphere is reconstructed by the focal centre $-e$, invisible to the template.*

This is the exact inverse of §4. There the ideal centre sat at infinity and the contact-set defect organised a *strict* $\mathcal{R}_X$; here the centre is an ordinary interior point with no boundary-contact structure to be "defective," it contributes most of $\mathcal{R}_X$, and reconstruction is *total*. The defect notion that runs the negative branch has no positive-curvature counterpart — there is nothing to detect, because everything off X is reconstructed (Remark 3.6 located the sign-flip that causes this).

Off constant curvature: what survives. Total reconstruction uses constant curvature twice — the cut point realises the diameter, in every direction at once (Remark 3.7's rigidity). Relaxing it, the conclusion persists under a spread hypothesis and the obstruction is exactly identified.

Theorem 6.3 (Spread suffices on any compact M; [5, Thm. 6.1]). *Let M be compact and $X \subseteq M$ closed and δ-dense with $2\delta < \operatorname{inj}(M)$. Then $\mathcal{R}_X = M \setminus X$. Under* $\sec \leq \Delta$ *the hypothesis is explicit via Klingenberg's estimate. For sparse X in variable positive curvature the sole obstruction is* cut-shielding *— a ray reaching the cut locus of its foot before any rival contact — and total reconstruction holds iff cut-shielding is absent [5, Prop. 7.1]. Cut-shielding genuinely occurs (Remark 6.6), so this conclusion is sharp.*

Theorem 6.4 (The two-geodesic axis removes the obstruction; [5, Thm. 8.3]). *Let M_X^g be the set of points with two distinct minimising geodesics to X (the cut locus of X) and $\mathcal{R}_X^g = \bigcup_{a \in M_X^g} B(a, r(a))$. Then for* every *compact M and closed X with $|X| \geq 2$, $\mathcal{R}_X^g = M \setminus X$, with no spread or curvature hypothesis; and $M_X^g = M_X$ on $\mathbb{R}^n$ and $\mathbb{S}^n$, so the sphere and Euclidean theories are unchanged. The gradient flow always rests at a critical point, which carries two minimising geodesics by definition; cut-shielding is precisely the case where these have a common endpoint, and the two-geodesic axis counts it as medial.*

Remark 6.5 (The scope of Theorem 6.4's proof). The reconstruction step — that the flow keeps p inside the moving free ball until it reaches a_∞ — is Lieutier's swallow property [10], proved in $\mathbb{R}^n$. The elementary form quoted in [5, Thm. 8.3], with the margin $r - d(p, \cdot)$ constant along each unique-foot arc, uses that the arc is the *minimising* geodesic from its foot, hence is rigorous exactly while the reach stays below that foot's cut distance: unconditional on $\mathbb{S}^n$ (cut distance $= \pi =$ diameter) and under the spread hypothesis of Theorem 6.3 ($2\delta < \operatorname{inj} M$). On a general compact M the reach can exceed an intermediate foot's cut distance, where the arc ceases to minimise and the elementary identity breaks; there $\mathcal{R}_X^g = M \setminus X$ rests on Lieutier's swallow extending past cut points (expected, and unconditional for generic or analytic metrics, but not delivered by the margin identity alone). We therefore read Theorem 6.4 as unconditional below the cut distance and as resting on the Lieutier property otherwise — "any compact M" in that qualified sense.

Both refinements stay on the finite-centre side: the flow lands at a *finite* critical point, and the only question is whether its two geodesics reach two *points* (then M_X, Theorem 6.3) or one (then only M_X^g, Theorem 6.4). The convexity detection of the negative branch never reappears, exactly as the trichotomy predicts.

Remark 6.6 (Sharpness: two-point reconstruction fails off constant curvature). The maximally sparse case settles, in the negative, whether Theorem 6.1 can hold for the two-*point* axis off constant curvature. Take $X = \{x_0, x_1\}$ on a prolate ellipsoid of revolution (intrinsic sec > 0 everywhere), with x_0 on the equator and x_1 at a pole. With $|X| = 2$ the two-point axis M_X is exactly the bisector $\{a : d(a, x_0) = d(a, x_1)\}$, so $q \in \mathcal{R}_X$ iff some bisector point a satisfies $d(a, q) < d(a, x_0)$ — a finite, decidable test. Evaluating it reveals a sharp transition in the eccentricity. (The solver uses Dijkstra geodesics on a dense point cloud; validated on the round sphere it returns distances within 1%, total reconstruction, and zero shielded points, confirming Theorem 6.1.) Reconstruction is total for mild prolateness (ratio of polar to equatorial semi-axes c small) and shielded for large eccentricity, the transition localised to $c^* = 2.6 \pm 0.1$ — the solver's 2.4% error cannot resolve it more finely. At $c = 3$ a robust *shielded* region near the antipodal pole occupies $\approx 9\%$ of the surface, with reconstruction deficit $\min_a \big[d(a, q) - r(a)\big] \approx 18\%$ of the diameter at its deepest point — the distance by which q lies beyond every medial ball. This deficit is a factor of about seven above the 2.4% discretisation error, and the shielded region persists and stabilises under mesh refinement (its area fraction rising slightly, from 7% to $\approx 9\%$ as the mesh doubles, rather than vanishing). The shielded points are precisely the cut points of x_0 that admit two or more minimising geodesics to x_0 while remaining strictly closer to x_0 than to x_1: meeting no rival contact, they lie in $M_X^g \setminus M_X$ and are reconstructed by the two-*geodesic* axis (Theorem 6.4) but by no two-point medial ball. Thus total reconstruction from M_X is restricted to constant curvature — and, more finely, to bounded eccentricity — whereas M_X^g is unconditional: cut-shielding is not a gap in the theory but the exact geometric obstruction separating the two axes, and the two-geodesic reformulation resolves it.

7 The unified picture

The phenomenon, in one line. A point escapes reconstruction by sending its nearest-point ray to an *end* of M with a single foot — so partial-versus-total is governed by **compactness**. *Given* partiality, what a contact set must do to admit a medial centre is governed by the sign of the Busemann Hessian: a convexity test where the centre is ideal ($\sec \le 0$), a contact count where it is finite ($\sec > 0$), Motzkin at the knife-edge. The two questions coincide on $\mathbb{H}^n$ and $\mathbb{S}^n$ but are independent (the paraboloid; the compact hyperbolic manifold).

The 2×2. The honest structure has *two* axes: compactness (rows: total versus partial) and the Hessian sign (columns: detection flavour, visible in the non-compact row).

	$\sec \le 0$ ($\mathrm{Hess}\, b \succeq 0$) ideal centre	$\sec \equiv 0$ ($\mathrm{Hess}\, b \equiv 0$) ideal at ∞	$\sec > 0$ ($\mathrm{Hess}\, b \preceq 0$) finite centre
non-compact (rays escape)	*partial*: Chebyshev/ Carnot defect ($\mathbb{H}^n$)	*partial*: Motzkin (Białożyt, $\mathbb{R}^n$)	*partial*: $\ge$ 2-count (paraboloid)
compact (no escape)	*total* $\mathcal{R}_X^g$ (cpt. hyperbolic)	*total* $\mathcal{R}_X^g$ (flat torus)	*total* $\mathcal{R}_X^g$ ($\mathbb{S}^n$; two-point too)

The columns are Theorem 3.2 with Propositions 3.5, 3.4 and Theorems 4.2, 5.1, 6.1; the rows are Theorem 3.1 with Theorem 6.4. The classical results sit on a diagonal — $\mathbb{H}^n, \mathbb{R}^n$ (non-compact) and $\mathbb{S}^n$ (compact) — which is why a first reading fuses the axes; the off-diagonal cells, the paraboloid and the compact space forms, are what pry them apart. Reading the non-compact row is the content of Theorem 6's detection-flavour generalisation: *not* one formula but a convexity question that trivialises into a count as the ideal centre descends to a finite one. The compact row is uniformly total *for the two-geodesic axis* $\mathcal{R}_X^g$ — but *not* for Białożyt's two-*point* axis M_X, the object of every column header: there cut-shielding (Remark 6.6) makes the compact-hyperbolic and the generic positively-curved cells *partial* for M_X, total only under a cut-locus rigidity ($\mathbb{S}^n$, where $M_X = M_X^g$). So the M_X-versus-M_X^g distinction is the genuine content of the compact row, not an aside; and Białożyt's own partiality was never about $\sec \le 0$ but about the non-compactness of $\mathbb{R}^n$.

What each branch leaves open. The two open problems are the two branches' own, and the trichotomy says they do not mix.

- *Negative branch (ideal centre).* Detection on non-CAT(0) horospheres: a Chebyshev-set problem in a Carnot group ($\mathbb{CH}^n$), or Motzkin's theorem in an Alexandrov space of indefinite curvature (pinched $\sec \le -a^2$) [4, §10]. This is a *convexity* question and exists only because the centre is ideal.
- *Positive branch (finite centre).* Non-detection for sparse X in *variable* positive curvature: cut-shielding *does* obstruct total reconstruction from the two-point axis (Remark 6.6: an explicit prolate-ellipsoid witness), so $\mathcal{R}_X = M \setminus X$ is special to constant curvature. This is a *cut-locus* question and exists only because the certificate has collapsed — there is no convexity content left, only the geometry of the cut locus — and the two-geodesic axis removes it entirely (Theorem 6.4). The open residue is quantitative, not existential: the eccentricity threshold below which even two points reconstruct everything.

The trichotomy thus also explains why the search for a positive-curvature mirror of the $\mathbb{CH}^2$ detection theorem fails: detection is an ideal-centre structure, and above zero there are no ideal centres [5, Thm. 3.4].

The natural sequel. Total reconstruction makes the *set* problem trivial above zero, which points to the *domain* reconstruction problem of Lieutier [10]: reconstruct an open $\Omega \subseteq \mathbb{S}^n$ from the medial axis of $\partial\Omega$ using only balls contained in Ω. One must state this carefully, on pain of repeating the conflation this note corrects. The confinement to Ω is an axis-(A) device — it bounds the inradius and so caps reach, as compactness does — not an axis-(B) curvature one; and interior reconstruction of a bounded Ω is itself total (Lieutier), with the *same* trivial defect as the sec > 0 count, since inscribed balls have finite equidistant centres. So the sequel *cannot* re-import the sec ≤ 0 Chebyshev theory: the trichotomy forbids a convexity defect on the finite-centre side. Its genuine content, if any, is the *Alexandrov geometry of* $\partial\Omega$ — the positively curved induced metric of the boundary — as a detection object in its own right, not a re-emergence of the horoball defect.

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